24/02/2024

Code A Phase 1

Aakash + BYJU'S

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005,
Ph.011-47623456

MM : 720

NCERT Booster Test Series-RM(P1)2324-T05A

Time : 200 Min.

Topics Covered:
Physics : Electric Charges & Fields, Electrostatic Potential and Capacitance, Current Electricity
Chemistry : Hydrocarbons, Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids
Botany : Sexual reproduction in Flowering Plants, Principles of Inheritance & Variation
Zoology : Human Reproduction, Reproductive Health

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15**.
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. The insulation property of air breaks down at 3×10^6 V/m. The maximum charge that can be given to a sphere of diameter 5 m is nearly

(1) 3×10^{-2} C

(2) 2×10^{-3} C

(3) 6×10^{-4} C

(4) 2×10^{-5} C

2. An electric dipole of dipole moment $\vec{P}$ is placed in a uniform electric field $\vec{E}$. Then

I. The resultant force on the dipole is zero.

II. The torque on the dipole is $\vec{P} \times \vec{E}$

III. The potential energy of the system is $-\vec{P} \cdot \vec{E}$

Which of the above statements is/are correct.

(1) Only I

(2) I and II

(3) I, II and III

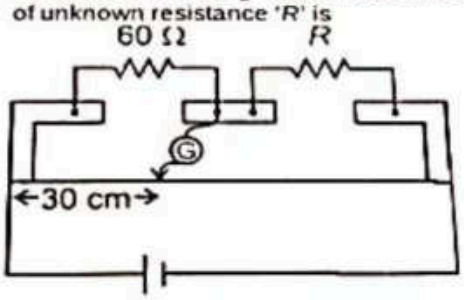
(4) I and III

@NEETXPR

29. If $\vec{J}$ and $\vec{E}$ are current density and electric field respectively inside a current carrying conductor at an instant then $\vec{J} \cdot \vec{E}$ is
(1) Positive
(2) Negative
(3) Zero
(4) May be positive or negative

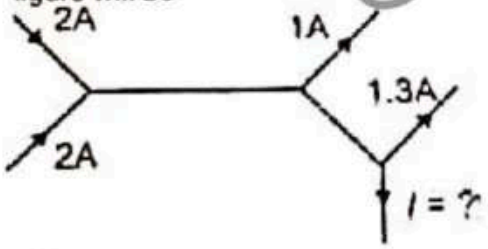
30. Two equal resistance when connected in series to a battery consume electric power of 100 W. If these resistance are now connected in parallel combination to the same battery, then electric power dissipated will be
(1) 800 W
(2) 25 W
(3) 200 W
(4) 400 W

31. In the shown set up of meter bridge, there is no deflection in the galvanometer. The value of unknown resistance 'R' is



(1) 60 Ω
(2) 120 Ω
(3) 140 Ω
(4) 160 Ω

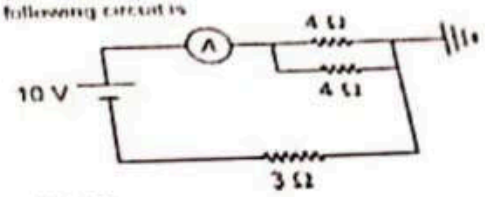
32. The current I in the part of a circuit shown in figure will be



(1) 1.7A
(2) 0.7A
(3) 4A
(4) Zero

33. When a negatively charged particle is taken from higher potential region to a lower potential region then its potential energy
(1) Decreases
(2) Increases
(3) Remains unchanged
(4) Becomes zero

34. The reading of ideal ammeter in the following circuit is



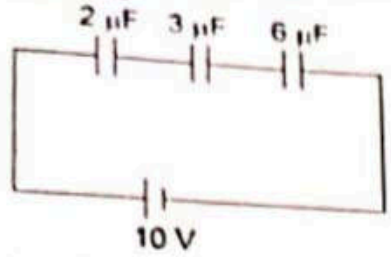
(1) 1 A
(2) 4 A
(3) 2.5 A
(4) 2 A

35. Kirchhoff's current law is based on the conservation of
(1) Momentum
(2) Charge
(3) Heat
(4) Force

@NEETXPR

SECTION-B

10. Three capacitors 2 μF, 3 μF and 6 μF are connected in series across a 10 V supply as shown in figure. Potential drop across 2 μF capacitor is



(1) 2 V
(2) 3 V
(3) 6 V
(4) 5 V

11. Electric potential at the centre of uniformly charged spherical shell, having charge q and radius R is

(1) $\frac{q}{4\pi\varepsilon_0 R^2}$
(2) $\frac{q}{4\pi\varepsilon_0 R}$
(3) $\frac{q}{8\pi\varepsilon_0 R^2}$
(4) $\frac{q}{8\pi\varepsilon_0 R}$

12. Two point charges $q_1 = 2\ \mu C$ and $q_2 = -1\ \mu C$ are kept at point $x = 0$ and $x = 6$ m respectively. Electric potential due to this system of charges will be zero at the points

(1) $x = 1$ m and $x = 3$ m
(2) $x = 2$ m and $x = 9$ m
(3) $x = 4$ m and $x = 12$ m
(4) $x = 5$ m and $x = 10$ m

13. In a region electric potential $V(x, y, z)$ is given as xyz. The electric field at point $P(1, -2, 3)$ is ??

(1) $12\hat{i} + 3\hat{j} + 2\hat{k}$
(2) $-6\hat{i} - 3\hat{j} + 2\hat{k}$
(3) $6\hat{i} - 3\hat{j} + 2\hat{k}$
(4) $-12\hat{i} + 3\hat{j} - 2\hat{k}$

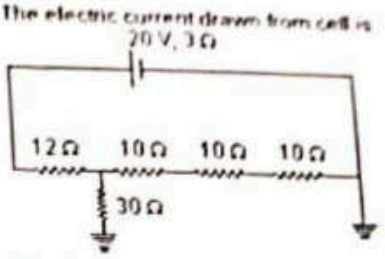
14. Dimensions of capacitance are

(1) $[M^{-1}L^{-1}T^4A^2]$
(2) $[ML^{-2}T^3A^2]$
(3) $[M^{-1}L^{-2}T^4A^2]$
(4) $[M^1L^{-3}T^3A^{-4}]$

15. The current flowing through a wire depends on time as $i = 3t^2 + 2t + 5$ A. The charge that flows through the cross section of the wire in time from $t = 0$ to $t = 2$ s is

(1) 22 C
(2) 20 C
(3) 18 C
(4) 5 C

16. The electric current drawn from cell is



(1) $\frac{2}{3}$ A
(2) $\frac{4}{9}$ A
(3) $\frac{4}{15}$ A
(4) Zero

17. On increasing the temperature of a conducting wire, its resistivity

(1) Decreases
(2) Increases
(3) Remain same
(4) First decreases and then increases

18. The emf of a cell is ε and its internal resistance is r. Its terminals are connected to a resistance R. The potential difference between the terminals is 1.6 V for $R = 4\ \Omega$, and 1.8 V for $R = 9\ \Omega$. Then

(1) $\varepsilon = 1$ V, $r = 1\ \Omega$
(2) $\varepsilon = 2$ V, $r = 1\ \Omega$
(3) $\varepsilon = 2$ V, $r = 2\ \Omega$
(4) $\varepsilon = 2.5$ V, $r = 0.5\ \Omega$

19. The number density of free electrons in a copper conductor is $9 \times 10^{27}\ m^{-3}$. How long does an electron take to drift from one end of a wire 3.0 m long to it's other end? (Cross-section area of wire is $3.0 \times 10^{-6}\ m^2$ and it is carrying a current of 3.0 A)

(1) 1.2 h
(2) 6.5 h
(3) 5.8 h
(4) 10.2 h

@NEETXPR

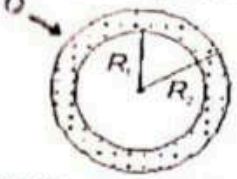
20. In a metre bridge experiment null point is obtained at 20 cm from one end of the wire when resistance X is balanced against another resistance Y. If X < Y then where will be the new position of the null point from the same end if one decides to balance a resistance of 4X against Y?
(1) 50 cm
(2) 80 cm
(3) 40 cm
(4) 70 cm

21. Two point charges A and B of +30 μC and −270 μC respectively are placed 10 cm apart as shown in the given figure. The point at which electric field intensity is zero is



(1) 2.5 cm right of A
(2) 2.5 cm left of B
(3) 5 cm left of A
(4) 5 cm right of A

22. If Q charge is distributed inside the material of a non-conducting shell of inner and outer radii R_1 and R_2 respectively, then,

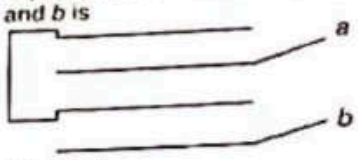


(1) Electric field inside shell must be zero
(2) Electric field inside shell may be zero
(3) Electric field inside shell is constant and non-zero
(4) Electric field inside shell is zero if Q is positive and non-zero if Q is negative

23. When an isolated metallic body is connected to earth then electrons from the earth flows into the body. This means that the body was
(1) Uncharged
(2) Positively charged
(3) Negatively charged
(4) May be (2) or (3)

24. Two identical pendulums A and B are suspended from the same point. The bobs A and B are given positive charges q_1 and q_2 respectively ($q_1 > q_2$). If A and B are making angles θ_1 and θ_2 with the vertical and T_1 and T_2 are tension in the string respectively, then select the correct option
(1) $\theta_1 > \theta_2$, $T_1 < T_2$
(2) $\theta_1 = \theta_2$, $T_1 > T_2$
(3) $\theta_1 = \theta_2$, $T_1 = T_2$
(4) $\theta_1 > \theta_2$, $T_1 = T_2$

25. Four identical metal plates are located in air at equal distance d from one another. The area of each plate is equal to A. The capacitance of the system between points a and b is



(1) $\frac{4\epsilon_0 A}{d}$
(2) $\frac{3\epsilon_0 A}{d}$
(3) $\frac{2\epsilon_0 A}{3d}$
(4) $\frac{3\epsilon_0 A}{2d}$

26. Six equal point charges q each are placed at six corners of a regular hexagon of side a. The work required to take a point charge q slowly from infinity to the centre of a hexagon is $\left[K = \frac{1}{4\pi\epsilon_0}\right]$
(1) Zero
(2) $\frac{Kq^2}{2a}$
(3) $\frac{3Kq^2}{a}$
(4) $\frac{6Kq^2}{a}$

27. Electric potential at any point on equatorial line of electric dipole is
(1) Directly proportional to r
(2) Inversely proportional to r
(3) Inversely proportional to r^2
(4) Zero

28. A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2 ms. The power delivered to the patient is
(1) 45 kW
(2) 90 kW
(3) 180 kW
(4) 360 kW

@NEETXPR

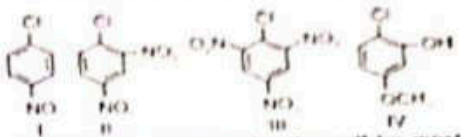
58. Given below are two statements
Statements (I): Vicinal dihalides on heating with zinc dust form an alkene
Statements (II): Dihalides in which two halogen atoms are attached to two adjacent carbon atoms are known as vicinal dihalides
In the light of above statements choose the correct option
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Statement I is incorrect and statement II is correct
(4) Statement I is correct and statement II is incorrect

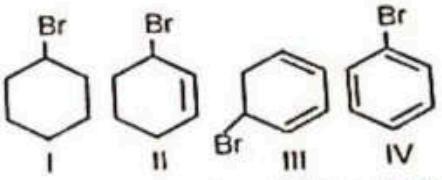
59. The halide ion which have maximum nucleophilicity in acetone is
(1) I^-
(2) Cl^-
(3) Br^-
(4) F^-

60. Chloroform on reaction with O_2 in presence of light gives
(1) CO_2
(2) $COCl_2$
(3) CO
(4) CCl_3-CHO

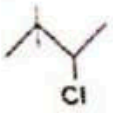
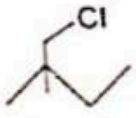
61. Which will react fastest with aqueous ethanol?
(1) $(CH_3)_3C-Br$
(2) $CH_2=CH-Br$
(3) $(CH_3)_2CH-Br$
(4) $CH_3-CH_2-CH_2-Br$

62. $CH_3CH_2Cl + NaI \xrightarrow{\text{Acetone}} CH_3CH_2I + NaCl$
The reaction is known as
(1) Finkelstein reaction
(2) Swarts reaction
(3) Hunsdiecker reaction
(4) HVZ reaction

63. 
Which of the above molecules will be most reactive towards aromatic nucleophilic substitution reaction?
(1) I
(2) II
(3) III
(4) IV

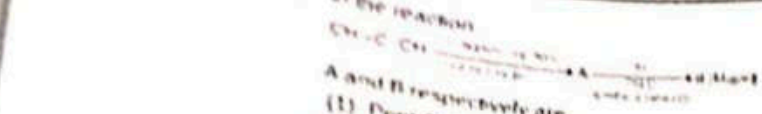
64. 
Correct reactivity order of above halides towards S_N1 reactions is
(1) III > II > I > IV
(2) II > I > III > IV
(3) IV > III > II > I
(4) III > IV > II > I

65. Which of the following has weakest C–Cl bond?
(1) Vinyl chloride
(2) Chlorobenzene
(3) 1-Bromo-2-chlorobenzene
(4) Chlorocyclopentane

66. Compound A(C_6H_{14}) gives three different monochlorides on photochemical chlorination. One of the monochlorides is inert to E–2 elimination, then that monochloride is
(1) 
(2) 
(3) 
(4) 

@NEETKPR

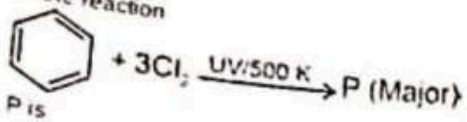
51. In the reaction



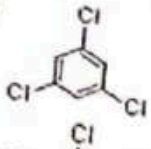
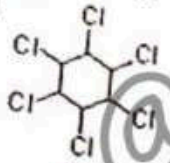
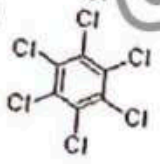
A and B respectively are
(1) Pent-2-yne, n-pentane
(2) But-2-yne, cis but-2-ene
(3) Pent-2-yne, trans-pent-2-ene
(4) Pent-2-yne, cis-pent-2-ene

52. Product of reductive ozonolysis of o-xylene is/are
(1) Glyoxal only
(2) Methyl glyoxal, glyoxal and 1, 2-dimethyl glyoxal
(3) Dimethyl glyoxal and glyoxal only
(4) Glyoxal and methyl glyoxal only

53. In the reaction

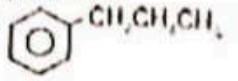
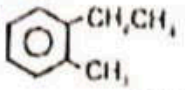
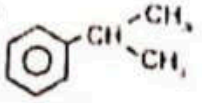
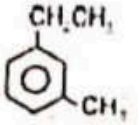


P is
(1)

(2)

(3)

(4)


54. In the following reaction



The product B is
(1)

(2)

(3)

(4)


55. Arrange the following in the order of decreasing acidic character and choose the correct option

ethane, acetylene, ethene
(i) (ii) (iii)

(1) (iii) > (i) > (ii)
(2) (i) > (ii) > (iii)
(3) (ii) > (iii) > (i)
(4) (ii) > (i) > (iii)

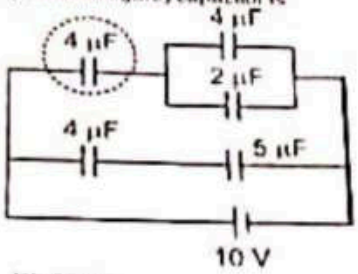
56. Which of the following cannot be obtained by the pyrolysis of C_5H_{14}?
(1) H_2
(2) C_6H_{12}
(3) C_7H_{16}
(4) CH_4

57. Elimination reaction of 2-Bromo-pentane to form Pent-2-ene is
(a) β-Elimination reaction
(b) Follows Zaitsev rule
(c) Dehydrohalogenation reaction
(d) Dehydration reaction

(1) (b), (c), (d) only
(2) (a), (b), (d) only
(3) (a), (b), (c) only
(4) (a), (c), (d) only

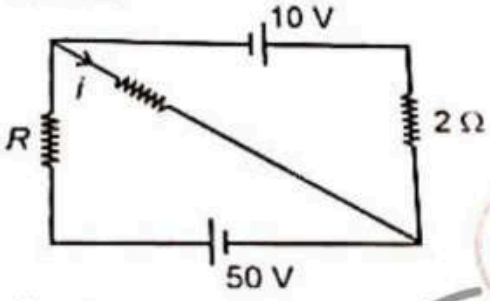
@NEETXDR

46. In the given circuit, the charge on 4 μF (circled in figure) capacitor is



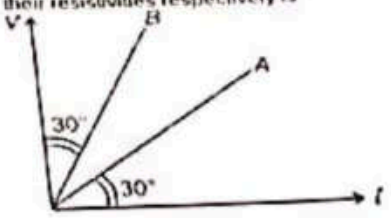
(1) 36 μC
(2) 24 μC
(3) 16 μC
(4) 90 μC

47. In given circuit, the current i will be zero when R is



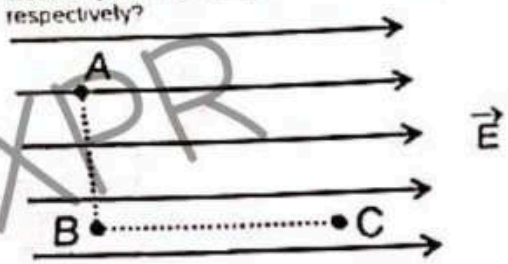
(1) 2 Ω
(2) 10 Ω
(3) 4 Ω
(4) 100 Ω

48. If two conductors A and B having identical dimensions but different material have V-I graphs as shown in figure, then the ratio of their resistivities respectively is



(1) 3 : 1
(2) 1 : 3
(3) 1 : 1
(4) 1 : 2

49. Figure shows three points A, B and C in a region of uniform electric field $\vec{E}$. The line AB is perpendicular and BC is parallel to the field lines. Then which of the following holds good. Where V_A, V_B and V_C represents the electric potential at points A, B and C respectively?



(1) $V_A = V_B = V_C$
(2) $V_A = V_B > V_C$
(3) $V_A = V_B < V_C$
(4) $V_A > V_B = V_C$

50. An electric bulb of rating 60 W, 220 V is connected across a voltage of 110 V. Power consumed by bulb is

(1) 60 W
(2) 12 W
(3) 15 W
(4) 30 W

CHEMISTRY

SECTION-A

@NEETXPR

3. Consider a uniform electric field $\vec{E} = 8 \times 10^3 \hat{i}$ N/C. What is the flux through a square of side length 10 cm, if normal to its plane makes an angle 60° with x-axis?

(1) $40 \frac{Nm^2}{C}$

(2) $80 \frac{Nm^2}{C}$

(3) $20 \frac{Nm^2}{C}$

(4) $10 \frac{Nm^2}{C}$

4. There are two charges $+2\,\mu C$ and $-3\,\mu C$. The ratio of mutual electrostatic forces acting on them will be

(1) 2 : 3

(2) 1 : 1

(3) 3 : 2

(4) 4 : 9

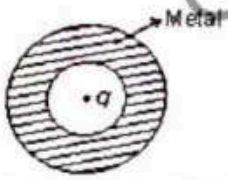
5. A : Two positively charged balls may attract each other.
R : When charge on ball is much larger than the other ball and they are placed close enough then they may attract.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

6. 

Charge q is placed at the centre of a conducting isolated shell, as shown in the figure. Distribution of charge on inner and outer surface of shell is

(1) Uniform on inner and non uniform on outer

(2) Uniform on both surfaces

(3) Non uniform on inner surface and uniform on outer

(4) Non uniform on both inner and outer surface

7. A short electric dipole is placed along x-axis at the origin. The electric field vector at a point P on y-axis is $\vec{E}_1$. The dipole is rotated by $\frac{\pi}{2}$ at its position. Now the electric field vector at point P is $\vec{E}_2$. Select the correct alternative.

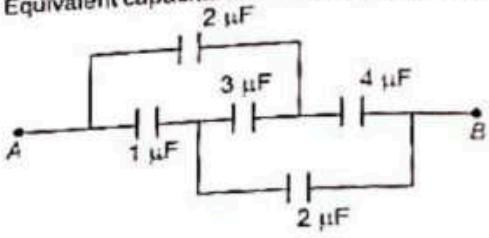
(1) $\vec{E}_2 = \vec{E}_1$

(2) $\left|\vec{E}_2 \times \vec{E}_1\right| = 0$

(3) $\left|\vec{E}_1\right| = 2\left|\vec{E}_2\right|$

(4) $\vec{E}_2 \cdot \vec{E}_1 = 0$

8. Equivalent capacitance between A and B is



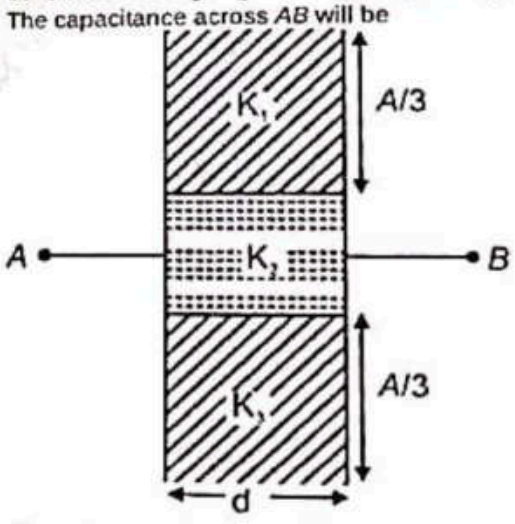
(1) 2 μF

(2) 0.5 μF

(3) 1 μF

(4) 1.5 μF

9. A parallel plate capacitor of plate area A and separation d filled with three dielectric materials as shown in figure. The dielectric constants are K_1, K_2 and K_3 respectively. The capacitance across AB will be



(1) $\frac{\varepsilon_0 A}{d}(K_1 + K_2 + K_3)$

(2) $\frac{\varepsilon_0 A}{3d}(K_1 + K_2 + K_3)$

(3) $\frac{3\varepsilon_0 A}{2d}(K_1 + K_2 + K_3)$

(4) $\frac{\varepsilon_0 A}{d}\left(\frac{K_1 K_2 K_3}{K_1 + K_2 + K_3}\right)$

@NEETXPR

36. Two particles of equal masses m and charge Q are placed at a distance of 20 cm. They do not experience any force. The value of $\frac{Q}{m}$ is
(1) Zero
(2) 1
(3) $\sqrt{\frac{4\pi\varepsilon_0}{G}}$
(4) $\sqrt{4\pi\varepsilon_0 G}$

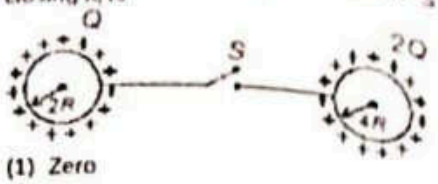
37. A charge Q is to be divided into two charges Q' and $Q - Q'$, such that electrostatic repulsion between them is maximum at a fixed distance R. The value $\frac{Q}{Q'}$ will appear to be
(1) 2
(2) 3
(3) $\frac{1}{3}$
(4) $\frac{1}{4}$

38. The electric field intensity in N/c due to a conducting infinite plate having surface charge density = 8.85 $\mu C/m^2$ at 10 cm is
(1) 10^{-6}
(2) 10^{6}
(3) 10^{-12}
(4) 10^{12}

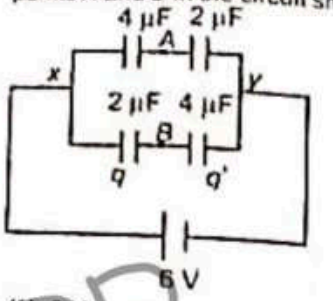
39. The electric potential at a distance $2R$ from the centre of a charged sphere having surface charge σ and radius R will be
(1) $\frac{R\sigma}{\varepsilon_0}$
(2) $\frac{R\sigma}{4\varepsilon_0}$
(3) $\frac{R\sigma}{2\varepsilon_0}$
(4) $\frac{2R\sigma}{\varepsilon_0}$

40. The kinetic energy of a particle is decreased by 20 J as it moves from a point at potential 50 V to a point at potential 100 V. Then charge on particle is
(1) +0.2 C
(2) −0.2 C
(3) +0.4 C
(4) −0.4 C

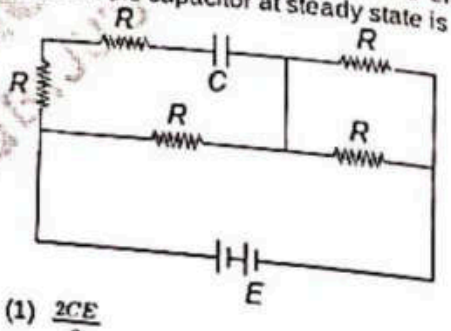
41. An electric field $\vec{E} = 10x\hat{i}$ exists in the space. Taking potential at (5 m, 8 m) to be zero, the potential at the origin is
(1) −100 V
(2) −150 V
(3) 175 V
(4) 125 V

42. Flow of charge through the switch S on closing it, is


(1) Zero
(2) Q
(3) $\frac{Q}{3}$
(4) $\frac{2Q}{3}$

43. What is the potential difference between points A and B in the circuit shown?


(1) 2 V
(2) 4 V
(3) 3 V
(4) 12 V

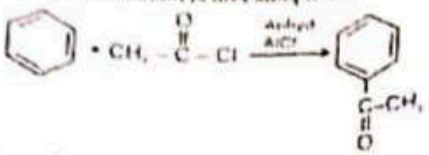
44. In the electrical network shown, the energy stored in the capacitor at steady state is


(1) $\frac{2CE}{3}$
(2) $\frac{CE^2}{9}$
(3) $\frac{2CE^2}{9}$
(4) $\frac{CE^2}{3}$

45. Two batteries A and B with e.m.f 10 V and 12 V are connected parallel across an external resistor of 3 Ω. If the internal resistance of A and B are 1 Ω and 3 Ω respectively, then the voltage across the external resistor is
(1) 7.6 V
(2) 8.4 V
(3) 10.5 V
(4) 12.2 V

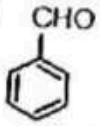
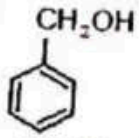
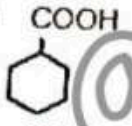
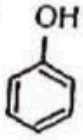
ONEETXER ALLEN + Digital

67. The given reaction is an example of



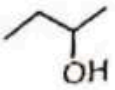
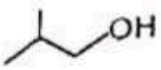
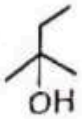
(1) Wurtz reaction
(2) Sandmeyer's reaction
(3) Friedel-Crafts reaction
(4) Fittig reaction

68. Which compound can give positive test with neutral $FeCl_3$?
(1) Ethyl alcohol
(2) Phenol
(3) Toluene
(4) Methyl alcohol

69. The compound that gives 2,4-DNP test is
(1) 
(2) 
(3) 
(4) 

70. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)
Assertion (A) : Phenol is more acidic than ethanol.
Reason (R) : Phenoxide ion is more stable than ethoxide ion.
In the light of above statements choose the correct answer.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

71. The correct order of boiling point of isomeric 1°, 2°, 3° alcohol is
(1) 1° > 2° > 3°
(2) 2° > 1° > 3°
(3) 2° > 3° > 1°
(4) 3° > 2° > 1°

72. Which alcohol will react fastest with Lucas reagent?
(1) 
(2) 
(3) 
(4) 

73. Cumene, on oxidation with O_2/air followed by acidification gives the following products
(1) Phenol only
(2) Acetone only
(3) Phenol and acetone
(4) Phenol and isopropyl alcohol

@NEETXDR

79. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R)

Assertion (A) : $ClCH_2COOH$ is more acidic than CH_3COOH

Reason (R) : Due to –I effect acidic nature of carboxylic acid decreases.

In the light of the above statements choose the correct answer from the options given below

(1) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**

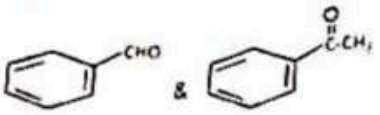
(2) **A** is true but **R** is false

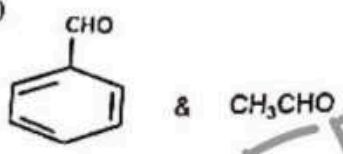
(3) **A** is false but **R** is true

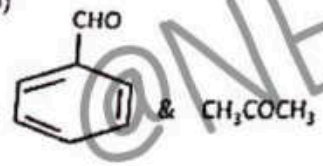
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**

80. Which of the following pairs can be distinguished by Fehling 's test?

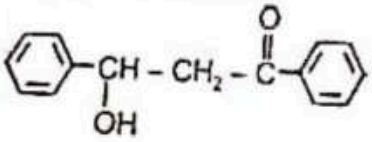
(1) HCHO & CH_3CHO

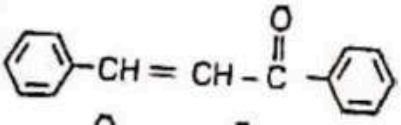
(2) 

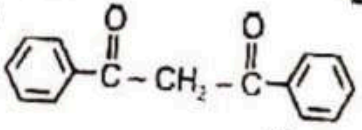
(3) 

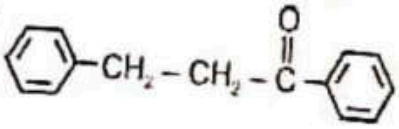
(4) 

81. When benzaldehyde is heated with acetophenone in presence of dilute NaOH then major product obtained is

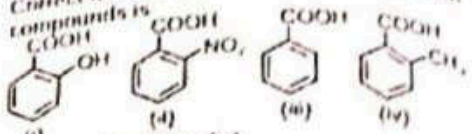
(1) 

(2) 

(3) 

(4) 

82. Correct order of acidic strength of the given compounds is



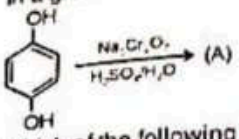
(1) (i) > (ii) > (iii) > (iv)

(2) (ii) > (iii) > (i) > (iv)

(3) (ii) > (iv) > (iii) > (i)

(4) (ii) > (i) > (iv) > (iii)

83. In a given reaction



Which of the following is correct about (A)?

(1) (A) contains 5 membered ring

(2) (A) does not show tautomerism

(3) (A) is carboxylic acid

(4) The given oxidation does not take place

@NEETXPR

84. Consider the following reaction sequence??

$CH_3COOH \xrightarrow{NH_3} A \xrightarrow{NaOBr} B \xrightarrow[H^+/H_2O]{(i)\ NaNO_2+HCl} D$ (Major)

Product D is

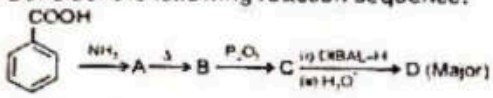
(1) CH_3CH_2OH

(2) CH_3CH_2Cl

(3) CH_3NO_2

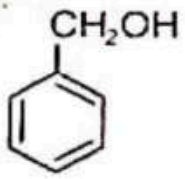
(4) CH_3OH

85. Consider the following reaction sequence.

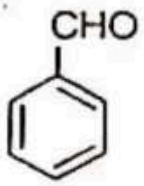


Major product D is

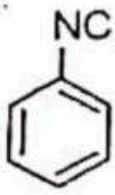
(1)



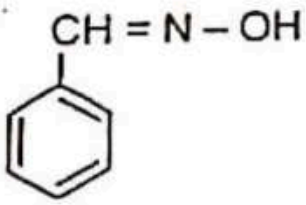
(2)



(3)



(4)



@NEETXPR

SECTION-B

86. Most reactive alkene towards dil. HCl is

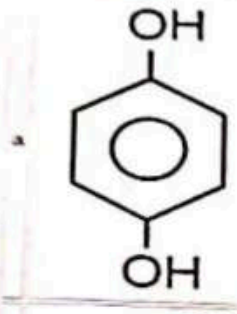
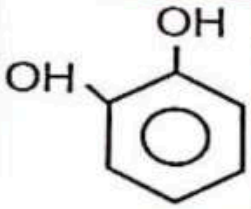
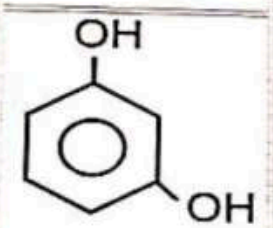
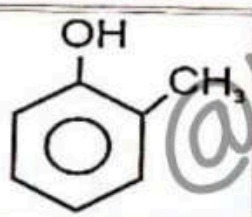
(1) Ethene

(2) Propene

(3) Butene

(4) 2,3-Dimethyl-2-butene

74. Match compounds in List I with their name in list II and choose the correct answer

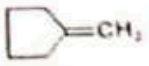
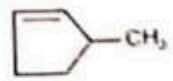
List I		List II	
a		(i)	Catechol
b		(ii)	Resorcinol
c		(iii)	Cresol
d		(iv)	Qainol

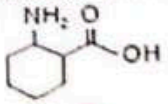
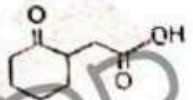
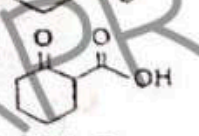
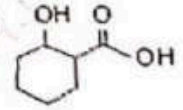
(1) a(iv), b(i), c(ii), d(iii)
(2) a(i), b(iv), c(ii), d(iii)
(3) a(iv), b(i), c(iii), d(ii)
(4) a(i), b(ii), c(iv), d(iii)

75. Formaldehyde on reaction with CH_3MgBr, followed by hydrolysis produces
(1) 1° alcohol
(2) 2° alcohol
(3) An acid
(4) An ester

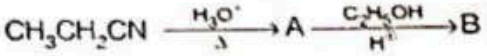
76. Consider the given reaction

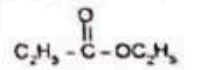


(Major) Product is
(1) 
(2) 
(3) 
(4) 

77. Which among the following compound decarboxylates most easily on heating?
(1) 
(2) 
(3) 
(4) 

78. Consider the following reaction sequence.



Compound B obtained is
(1) $CH_3CH_2NH_2$
(2) $CH_3CH_2 - OC_2H_5$
(3) CH_3CH_2COOH
(4) 

@NEETXPR

99. $C_2H_5NC + H_2O \xrightarrow{H^+} A + B$
A and B are
(1) $C_2H_5NHCH_3$ and CH_3NH_2
(2) $C_2H_5NH_2$ and HCOOH
(3) CH_3NH_2 and CH_3COOH
(4) C_2H_5OH and N_2

100. Which acid is majorly present in lemon juice?
(1) Malic acid
(2) Maleic acid
(3) Citric acid
(4) Stearic acid

BOTANY

SECTION-A

101. Skin colour in humans is an example of
(1) Polygenic inheritance
(2) Co-dominance
(3) Incomplete dominance
(4) Qualitative inheritance

102. Pericarp of a fruit is
(1) Wall of ripened ovary
(2) Integument of mature ovule
(3) Fleshy corolla
(4) Ripened thalamus

103. The **correct** sequence of different parts of an ovule from outside to inside is
(1) Nucellus, Integument, Embryo sac
(2) Integument, Nucellus, Embryo sac
(3) Nucellus, Funicle, Chalaza
(4) Integument, Funicle, Nucellus

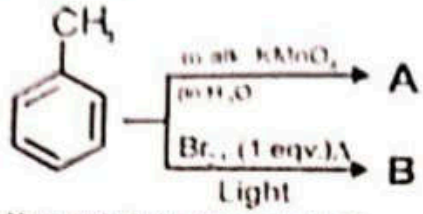
104. Who amongst the following was **not** involved in the rediscovery of the Mendel's result on the inheritance of characters?
(1) de Vries
(2) Correns
(3) von Tschermak
(4) Boveri

105. Select the **incorrect** statement w.r.t. tapetum
(1) It is the innermost layer of anther wall
(2) It nourishes the developing pollen grains
(3) Its cells possess dense cytoplasm
(4) Its cells generally have more than one haploid nucleus

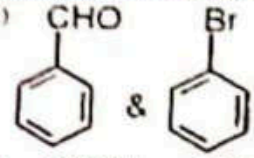
106. Mature female gametophyte of angiosperms is generally
(1) 7-celled & 7-nucleated structure
(2) 8-celled & 7-nucleated structure
(3) 7-celled & 8-nucleated structure
(4) 3-celled & 3-nucleated structure

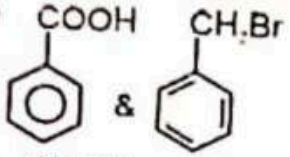
107. Characteristics of a flower are as
a. Nectaries absent
b. Flowers packed into inflorescence.
c. Well exposed stamens.
d. Light weighted, non-sticky pollens.
On the basis of above features, flower must be
(1) Hydrophilous
(2) Anemophilous
(3) Entomophilous
(4) Cleistogamous

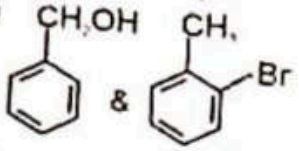
@NEETXPR

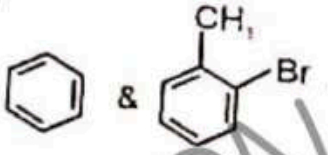
87. 

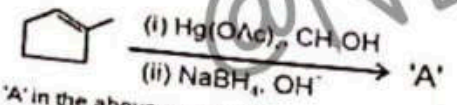
Major products A & B respectively are

(1) 

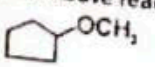
(2) 

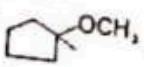
(3) 

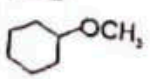
(4) 

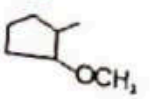
88. 

'A' in the above reaction is

(1) 

(2) 

(3) 

(4) 

89. Given below are the two statements

Statement I : Alkyl fluorides on reduction with zinc and dilute HCl gives alkanes.

Statement II : Wurtz reaction is done in presence of dry ether.

In light of the above statements, choose the **correct** answer

(1) Statement I is correct but statement II is incorrect.

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

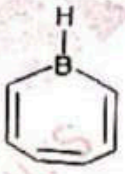
(4) Both statement I and statement II are incorrect

90. Which of the following is aromatic in nature?

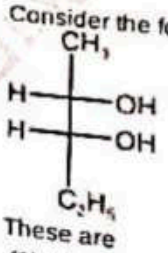
(1) 

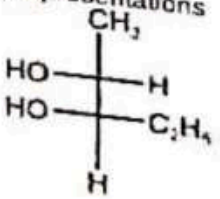
(2) 

(3) 

(4) 

91. Consider the following representations





These are

(1) Identical

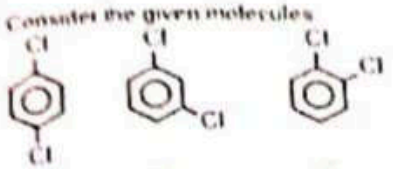
(2) Enantiomers

(3) Conformational isomers

(4) Diast...

@NEETXDR

92. Consider the given molecules



(I) (II) (III)

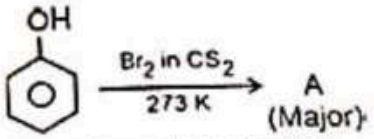
The correct order of melting point will be

(1) I > III > II
(2) III > II > I
(3) II > III > I
(4) I > II > III

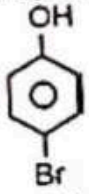
93. $C_2H_5OH \xrightarrow{PCl_5} X + POCl_3 + HCl$

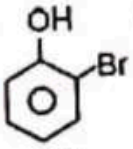
Compound X is

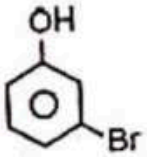
(1) CH_3Cl
(2) C_2H_5Cl
(3) CH_3OCH_3
(4) $C_2H_5OC_2H_5$

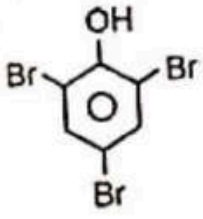
94. 

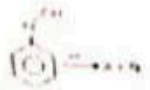
A is major product. Identify A.

(1) 

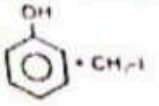
(2) 

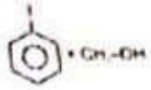
(3) 

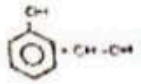
(4) 

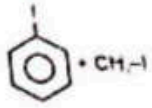
95. 

The products A and B are?

(1) 

(2) 

(3) 

(4) 

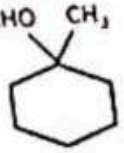
96. When glucose undergoes fermentation in the presence of zymase, it gives

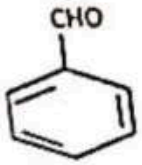
(1) Ethanoic acid
(2) Methyl alcohol
(3) Ethyl alcohol
(4) Formic acid

97. Which of the following is not a dibasic acid?

(1) Oxalic acid
(2) Picric acid
(3) Malonic acid
(4) Terephthalic acid

98. Cannizzaro reaction is not given by

(1) 

(2) 

(3) HCHO
(4) CH_3CHO

@NEETXPR

(3) CH_3NH_2 and CH_3COOH
(4) C_2H_5OH and N_2

100. Which acid is majorly present in lemon juice?
(1) Malic acid
(2) Maleic acid
(3) Citric acid
(4) Stearic acid

CC-020 CC-020 CC-020 CC-020

BOTANY

SECTION-A

101. Skin colour in humans is an example of
(1) Polygenic inheritance
(2) Co-dominance
(3) Incomplete dominance
(4) Qualitative inheritance

102. Pericarp of a fruit is
(1) Wall of ripened ovary
(2) Integument of mature ovule
(3) Fleshy corolla
(4) Ripened thalamus

103. The **correct** sequence of different parts of an ovule from outside to inside is
(1) Nucellus, Integument, Embryo sac
(2) Integument, Nucellus, Embryo sac
(3) Nucellus, Funicle, Chalaza
(4) Integument, Funicle, Nucellus

CC-020 CC-020

104. Who amongst the following was **not** involved in the rediscovery of the Mendel's result on the inheritance of characters?
(1) de Vries
(2) Correns
(3) von Tschermak
(4) Boveri

105. Select the **incorrect** statement w.r.t. tapetum
(1) It is the innermost layer of anther wall
(2) It nourishes the developing pollen grains
(3) Its cells possess dense cytoplasm
(4) Its cells generally have more than one haploid nucleus

106. Mature female gametophyte of angiosperms is generally
(1) 7-celled & 7-nucleated structure
(2) 8-celled & 7-nucleated structure
(3) 7-celled & 8-nucleated structure
(4) 3-celled & 3-nucleated structure

CC-020

107. Characteristics of a flower are as
a. Nectaries absent
b. Flowers packed into inflorescence.
c. Well exposed stamens.
d. Light weighted, non-sticky pollens.
On the basis of above features, flower must be
(1) Hydrophilous
(2) Anemophilous
(3) Entomophilous
(4) Cleistogamous

Aakash +DIGITAL @NEETXDR

... which of the given plants autogamy as well as geitonogamy is prevented?

(1) Castor
(2) Maize
(3) Papaya
(4) Pea

109. According to 'Law of segregation'

(1) Two factors of a character get mixed up in F_2 generation
(2) During gamete formation, factors of a character get mixed up
(3) Factors of a character separate from each other during gamete formation
(4) Out of two factors for each character, dominant will be able to express in F_1 generation

110. "In 60% of angiosperms pollination takes place at A celled stage of a pollen, having B shaped generative cell that floats in the cytolasm of C cell".
Complete the above statement by choosing the **correct** option for A, B and C.

	A	B	C
(1)	Two	Irregular	Vegetative
(2)	Two	Spindle	Vegetative
(3)	Three	Irregular	Vegetative
(4)	Three	Spindle	Gametic

(1) (1)
(2) (2)
(3) (3)
(4) (4)

111. Select the odd one out **w.r.t.** Mendelian disorders

(1) Sickle-cell anaemia
(2) Down's syndrome
(3) Thalassemia
(4) Colour blindness

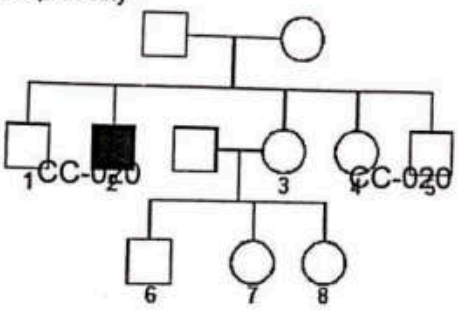
112. In a family, father is having B blood group and mother is of A blood group. If their children show 50% probability of AB blood group, it indicates that

(1) Both father and mother are homozygous
(2) Both mother and father are heterozygous
(3) Either of the parent is heterozygous
(4) Mother has a genotype $I^A I^A$ and father has genotype $I^A I^B$

113. Thalassemia is a/an

(1) Autosomal dominant disorder
(2) Autosomal recessive disorder
(3) X-linked dominant disorder
(4) X-linked recessive disorder

114. Study the given pedigree. Identify the possible genotype for 3rd and 4th individuals respectively



(1) 3-Aa, 4-aa
(2) 3-AA, 4-aa
(3) 3-aa, 4-Aa
(4) 3-Aa, 4-AA

115. Both chasmogamous and cleistogamous flowers are produced by

(1) Pea and *Zostera*
(2) Bean and *Vallisneria*
(3) Maize and bamboo
(4) *Oxalis* and *Commelina*

116. Select the **odd** one w.r.t. polygenic inheritance.

(1) It demonstrates segregation and assortment of genes controlling quantitative traits
(2) Extreme phenotypes are rare and the intermediate ones are more frequent
(3) Dominant alleles do not show cumulative effect
(4) Common examples are skin colour in man and human height

117. Genotype of a true breeding tall pea plant can be written as

(a) TT
(b) Tt
(c) tt

(1) (a) only
(2) (b) only
(3) (c) only
(4) Both (a) & (b)

CC-020 CC-020 CC-020 CC-020 CC-020 CC-020 CC-020
@NEETXPR Aakash BYJU'S

118. **Assertion (A)** : Embryo development precedes endosperm development.
Reason (R) : Early stages of embryogeny are different in both monocots and dicots though seeds are similar.
In the light of the above statements, Select the **correct** option

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

CC-020 CC-020

119. Read the following statements A and B and select the **correct** option
A. Fruit developed without fertilization is called false fruit.
B. In Lupine Seed viability is less than 2000 years.

(1) Only statement 'A' is incorrect

(2) Only statement 'B' is incorrect

(3) Both Statements A and B are incorrect

(4) Both Statements A and B are correct

120. Select the **correct** set of parents those can produce child with blood group 'O'.
a. $I^Ai \times I^Ai$
b. $I^Bi \times i^AI^A$
c. $ii \times I^Ai$
d. $I^BI^B \times I^Bi$

(1) a and c

(2) b and d

(3) a and d

(4) b and c

121. Who united the knowledge of chromosomal segregation with Mendelian principle and called it the chromosomal theory of inheritance?

(1) T.H. Morgan

(2) Walter Sutton

(3) Theodore Boveri

(4) Gregor Mendel

CC-020 CC-020

122. Four genes a, b, c and d are present on a chromosome. Percentage of recombination between a and b = 20%; a and d = 2%, c and b = 8% and d and b = 18%. What will be the sequence of genes on the chromosome?

(1) adcb

(2) cdab

(3) dcab

(4) acbd

123. Two plants of garden pea with the genotype RrYy (Round, Yellow seeds) are crossed with each other. What is the probability for the offsprings to carry only one of the dominant characters?

(1) $\frac{1}{16}$

(2) $\frac{5}{16}$

(3) $\frac{6}{16}$

(4) $\frac{3}{16}$

124. Which of the following features are **true** for synergids?
(a) Diploid cells
(b) Cells of egg apparatus
(c) Produced by microsporogenesis
(d) Contains filiform apparatus

(1) (a) and (b)

(2) (b) and (d)

(3) (a) and (c)

(4) (c) and (d)

CC-020 CC-020

125. 'X-body' observed by Henking in few insects is

(1) Received by all their sperms after spermatogenesis

(2) The X-chromosome of insects

(3) Present in only 50% of the eggs in grasshoppers

(4) Found in only those species which have XY type of sex determination

126. Who proposed the method of a graphical representation to calculate the probability of all possible genotypes of offsprings in a genetic cross?

(1) R.C. Punnett

(2) G. J. Mendel

(3) Bateson

(4) Johannser

127. (i) of pollen grain has (ii) which is not degraded by any enzyme.
Select the **correct** option to fill in the blanks (i) and (ii)

(1) (i)-Exine, (ii) – Lignin

(2) (i)-Exine, (ii) - Pectin

(3) (i)-Exine, (ii) - Sporopollenin

(4) (i)-Intine, (ii) - Chitin

CC-020 CC-020

Aakash +Byju's
@NEETXPR

How many of the following features of fruit flies make them suitable experimental material for genetic studies?
(a) Their short life span.
(b) They could be easily grown on simple synthetic medium.
(c) Male and female flies are not easily distinguishable.
(d) Their hereditary variations are seen only by using very high power microscopes.
(1) One
(2) Two
(3) Three
(4) Four

CC-020 CC-020

129. Pollen grain of angiosperms represents
(1) Male gamete
(2) Microsporangium
(3) Male gametophyte
(4) Megaspore

130. **Assertion :** In sickle cell anaemia, the shape of RBC changes from biconcave disc to elongated sickle like structure.
Reason : The mutant haemoglobin molecule undergoes polymerization under low oxygen tension causing the change in shape of RBC.
In the light of the above statements, select the **correct** option
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

131. Removal of anther from bisexual flower buds before the anther dehisces is called
(1) Bagging
(2) Anthesis
(3) Emasculation
(4) Artificial hybridisation

132. How many types of gametes are possible from genotype AaBBCcDd?
(1) 3
(2) 16
(3) 8
(4) 4

CC-020 CC-020

133. How many nuclei are involved in double fertilisation in an angiospermic plant?
(1) Two
(2) Three
(3) Six
(4) Five

134. Mark the **odd** one w.r.t. characteristics of wind pollinated flowers.
(1) Flowers packed into an inflorescence
(2) Large feathery stigma
(3) Well exposed stamens
(4) Pollen grains surrounded by mucilaginous covering

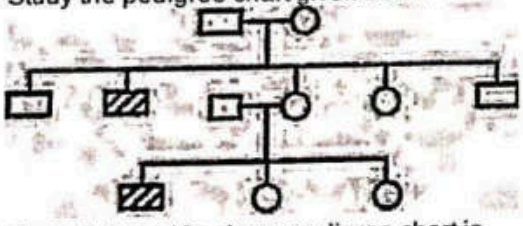
135. XO type of sex-determination in which the males have only one X-chromosome besides the autosomes whereas females have a pair of X-chromosomes is shown by
(1) Grasshopper
(2) *Drosophila*
(3) Humans
(4) Honey bees

@NEETXPR

SECTION-B

CC-020 CC-020

136. Study the pedigree chart given below.



The trait traced in above pedigree chart is
(1) X-linked dominant
(2) Y-linked
(3) Autosomal dominant
(4) Autosomal recessive

CC-020 CC-020

137. α-thalassemia is associated with chromosome number
(1) 16
(2) 11
(3) 9
(4) 22

138. Different forms of a gene are called
(1) Traits
(2) Phenotypes
(3) Complementary genes
(4) Alleles

139. Select the **wrong** match.
(1) Sickle cell anaemia - Substitution of valine by glutamic acid
(2) Haemophilia - Gene present on sex chromosome
(3) Thalassemia - Production of α or β globin chains is affected
(4) Phenylketonuria - An inborn error of metabolism

140. Which of the following trait **cannot** express itself in heterozygous condition?
(1) Wrinkled seed shape
(2) Yellow seed colour
(3) Violet flower colour
(4) Inflated pod shape

141. Trisomy of sex chromosome results in
(1) Down's syndrome
(2) Klinefelter's syndrome
(3) Turner's syndrome
(4) Cystic fibrosis

142. How many of the following have non-albuminous seeds?
Wheat, Maize, Barley, Castor, Pea, Groundnut
(1) 2
(2) 3
(3) 4
(4) 6

143. Which one is **correct** sequence of development of zygote into embryo?
(1) Pro-embryo → Heart shaped embryo → globular embryo → Mature embryo
(2) Pro-embryo → Globular embryo → Heart shaped embryo → Mature embryo
(3) Heart shaped embryo → Pro-embryo → Globular embryo → Mature embryo
(4) Globular embryo → Heart shaped embryo → Pro-embryo → Mature embryo

144. Complete the given statements by choosing the **correct** option for A, B, C and D.
a. Among the insects, ___A___ are the dominant biotic pollinators.
b. ___B___ and ___C___ provide safe places for egg laying as the floral rewards.
c. Insects which consume pollen or nectar without bringing about pollination are called ___D___.
(1) A. Butterflies, B. *Aquilegia*, C. *Hydrilla*, D. Pollen robbers
(2) A. Bees, B. *Yucca*, C. *Amorphophallus*, D. Pollen robbers
(3) A. Beetles, B. *Yucca*, C. *Amorphophallus*, D. Pollen robbers
(4) A. Bees, B. *Aquilegia*, C. *Pronuba*, D. Pollen robbers

145. In a clinical diagnosis a female 'A' was identified with absence of secondary sexual characters and has rudimentary ovaries. Which of the given statements can be **true** for the female 'A'?
(1) The chromosome complement of the female cannot be 44 + XO
(2) The female is suffering from a sex recessive linked disorder
(3) Mother of this female may be producing abnormal egg (22 + O) and father producing normal sperm (22 + X)
(4) This female may pass this disorder to her daughters

146. Find the **correct** match w.r.t. seed.
(1) Seed coat – Represents integument of ovary
(2) Micropyle – Represents pore of ovary through which pollen tube enters
(3) Hilum – Represents the point at which body of ovule was attached with funicle
(4) Endosperm – Represents product of syngamy

147. Haplodiploid method of sex determination is found in
(1) Butterfly
(2) Honey bee
(3) Grasshopper
(4) Fruit fly

148. The recessive allele
(1) May produce non functional enzyme
(2) Is always unmodified allele
(3) Always produces normal enzyme
(4) Can express itself in heterozygous condition

CC-020 CC-020 CC-020 CC-020 CC-020 CC-020 CC-020 NEETXPRO Aakash +Byju's

...ead the following statements and choose ... as true (T) or false (F)

a. In artificial hybridisation it is important to make sure that only the desired pollen grains are used for pollination

b. Artificial hybridisation is one of the major approaches of crop improvement programme.

c. If female parent produces unisexual flowers, there is no need for emasculation.

d. Filiform apparatus guides the entry of pollen tube into ovary.

	a	b	c	d
(1)	T	F	T	F
(2)	T	T	F	F
(3)	F	T	F	T
(4)	T	T	T	F

(1) (1)
(2) (2)
(3) (3)
(4) (4)

CC-020 CC-020 @NEETXPR

150. Correct set of dominant traits w.r.t. flowers in pea plant, which were studied by Mendel are

(1) Violet colour and axial position
(2) Pink colour and terminal position
(3) White colour and axial position
(4) Violet colour and terminal position

Aakash BYJU'S

ZOOLOGY

SECTION-A

151. Which one of the following is not a part of the female reproductive system in humans?

(1) A pair of ovaries
(2) A pair of mammary glands
(3) A pair of uterus
(4) A pair of oviducts

CC-020 CC-020

152. The narrow part of the fallopian tube closest and linked to the uterus is

(1) Infundibulum
(2) Fimbriae
(3) Ampulla
(4) Isthmus

153. Select the **correct** sequence representing the route of sperms transport through the human male reproductive system.

(1) Seminiferous tubules → Vasa efferentia → Rete testis → Epididymis
(2) Rete testis → Seminiferous tubules → Vasa efferentia → Epididymis
(3) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis
(4) Vas deferens → Rete testis → Vasa efferentia → Epididymis

CC-020 CC-020

154. Which of the following is **not** formed as a result of meiosis?

(1) Primary oocyte
(2) Secondary oocyte
(3) First polar body
(4) Second polar body

155. Birth canal is formed by
(1) Cervix only
(2) Vagina only
(3) Uterus along with cervical canal
(4) Cervical canal along with vagina

156. Choose the **incorrect** match w.r.t. human female reproductive system.
(1) Mons pubis – A cushion of fatty tissue covered with skin and pubic hair
(2) Labia majora – Fleshy folds of tissue extending down from mons pubis
(3) Ampulla of oviduct – Site of fertilization
(4) Mammary lobes – 40-50 in each breast

CC-020 CC-020

157. The structure that comes out of the testis open into the structure located along the posterior surface of each testis is
(1) Vasa efferentia
(2) Epididymis
(3) Vas deferens
(4) Seminiferous tubule

158. The structure that extends through the penis is
(1) Urethra
(2) Urinary bladder
(3) Seminal vesicle
(4) Testis

159. Match column I with column II and choose the option with all **correct** matches.

	Column I		Column II
a.	Insemination	(i)	Delivery of the baby
b.	Implantation	(ii)	Embryonic development
c.	Gestation	(iii)	Attachment of blastocyst to uterine wall
d.	Parturition	(iv)	Transfer of sperms into the female genital tract

(1) a(i), b(ii), c(iii), d(iv)
(2) a(iv), b(iii), c(ii), d(i)
(3) a(ii), b(i), c(iv), d(iii)
(4) a(iii), b(iv), c(i), d(ii)

CC-020 CC-020

160. Total number of sperms and ova produced by 100 primary spermatocytes and 100 secondary oocytes will be respectively
(1) 100 sperms and 100 ova
(2) 200 sperms and 200 ova
(3) 400 sperms and 100 ova
(4) 400 sperms and 200 ova

161. In a human female, menstruation can be deferred by the administration of
(1) Only FSH
(2) Only LH
(3) Combination of LH and FSH
(4) Combination of estrogen and progesterone

162. The layer of the wall of the uterus which exhibits strong contraction during delivery of the baby is
(1) Endometrium
(2) Myometrium
(3) Perimetrium
(4) Perisalpinx

CC-020 CC-020

163. If a female shows 36 days menstrual cycle, then what would be the approximate day of maximum LH level?
(1) 14th day
(2) 18th day
(3) 22nd day
(4) 28th day

164. A 25-year old healthy female who was otherwise having regular menstrual cycles suddenly skips menstruation for two consecutive cycles. The most likely cause of this is
(1) Degradation of corpus luteum
(2) Fertilization of ovum
(3) Oversecretion of FSH
(4) Onset of menopause

165. Which of the following is an **incorrect** match regarding embryonic development at various months of pregnancy in humans?
(1) End of 1st month → Heart is formed
(2) End of first trimester → Eye-lids separate and eyelashes are formed
(3) During 5th month → Appearance of hair on the head
(4) End of 2nd month → Formation of limbs and digits

CC-020 CC-020

@NEETKPR

Match the following and select the correct option

	Column I		Column II
(a)	Menstrual phase	(i)	Release of ovum
(b)	Follicular phase	(ii)	Fully mature Graafian follicle
(c)	Ovulatory phase	(iii)	Corpus luteum
(d)	Luteal phase	(iv)	Breakdown of endometrium

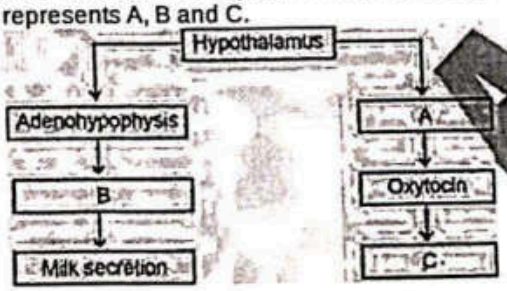
(1) a(iv), b(ii), c(i), d(iii)
(2) a(iv), b(i), c(ii), d(iii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(iv), b(iii), c(ii), d(i)

CC-020 CC-020

167. An ART in which sperm is directly injected into ovum is termed as
(1) GIFT
(2) ZIFT
(3) ICSI
(4) ET

168. Which hormone peaks twice during menstrual cycle?
(1) Progesterone, an ovarian hormone
(2) FSH, a pituitary hormone
(3) LH, a pituitary hormone
(4) Estrogen, an ovarian hormone

169. Identify A, B and C in the following flowchart and choose the option which correctly represents A, B and C.



	A	B	C
(1)	Anterior pituitary	Prolactin	Parturition
(2)	Neurohypophysis	Prolactin	Milk ejection
(3)	Pars distalis	Progesterone	Milk secretion
(4)	Pars nervosa	Milk ejection	Parturition

(1) (1)
(2) (2)
(3) (3)
(4) (4)

CC-020 CC-020

170. Use of how many of the following can alter/interfere with the libido of the user?

Diaphragm, Implants, Cu7, Oral contraceptive pills, Foams

Select the correct option
(1) Five
(2) Four
(3) Three
(4) Zero

171. Though all persons are vulnerable to STIs, their incidences are very high among persons belonging to which age group?
(1) 9-12 years
(2) 40-50 years
(3) 15-24 years
(4) 25-40 years

CC-020 CC-020

172. Choose the **wrong** statement w.r.t. medical termination of pregnancy (MTP).
(1) Nearly 45 to 50 million MTPs are performed each year all over the world
(2) It is mainly performed for the purpose of controlling population growth
(3) MTPs are considered relatively safe if performed during first trimester of pregnancy
(4) Total number of MTPs per formed in a year, accounts to 1/5 th of the total number of conceived pregnancies in that year

173. Use of which of the following contraceptive methods inhibits ovulation and implantation?
(1) Oral contraceptive pills
(2) Tubectomy
(3) Non-medicated IUDs
(4) Copper releasing IUDs

174. Choose the correct statement w.r.t. contraceptive that provides additional benefit of protecting the users from contracting STIs.
(1) It is made up of thin rubber/latex sheath that is used to cover the penis in males.
(2) It is irreversible.
(3) It prevents conception by allowing the entry of sperms through the cervix.
(4) It increases phagocytosis of sperms in the uterus.

CC-020 CC-020

175. Diaphragms, cervical caps, and vaults are used to cover
(1) Vagina
(2) Cervix
(3) Penis
(4) Fundus of uterus

@NEETXPR

176. The mitotic division starts, as the zygote moves through the isthmus of the oviduct is known as
(1) Gametogenesis
(2) Implantation
(3) Cleavage
(4) Lactation

177. Which of the following occurs after degeneration of corpus luteum?
(1) Rapid decrease in secretion of LH
(2) Rise in level of progesterone
(3) Disintegration of endometrium
(4) Rapid decrease of FSH

178. All of the following are true w.r.t. 'Saheli', **except**
(1) Inhibits implantation as well as ovulation
(2) Contains non-steroidal preparation
(3) Developed at Lucknow, India
(4) Once a week pill

179. Which of the following statements is **incorrect** regarding test tube baby?
(1) Early embryo (with upto 8 blastomeres) is transferred into the fallopian tube
(2) Embryo with more than 8 blastomeres is transferred into the uterus
(3) ICSI method is also considered as a type of IVF
(4) Ovum gets implanted in the fallopian tube and further development occurs in uterus

180. Which of the following is reusable contraceptive made of rubber and gives privacy to user?
(1) Condoms
(2) Lippes loop
(3) Cervical caps
(4) CuT

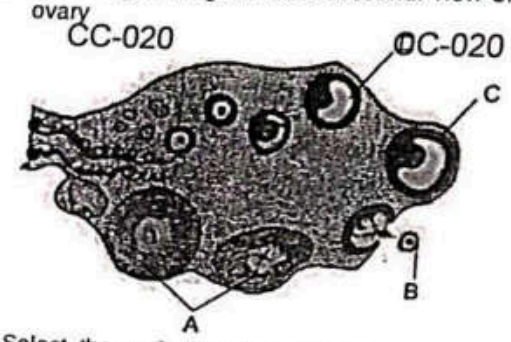
181. How many diseases given in the box below are considered as venereal disease?

Gonorrhoea, Measles, Hepatitis B, Diphtheria, Syphilis, Genital herpes, Genital warts, Dengue

Select the correct option
(1) Four
(2) Five
(3) Seven
(4) Three

182. All the given statements are correct regarding ZIFT, **except**
(1) Ova from wife / donor (female) and sperms from husband / donor (male) are collected
(2) Zygote is formed under simulated conditions in the laboratory
(3) The zygote with upto 8 blastomeres could be transferred into fallopian tube
(4) The zygote with more than 8 blastomeres stage could be transferred into fallopian tube

183. Following is diagrammatic sectional view of ovary



Select the option which gives the correct identification of A, B, C and D with function/characteristics.
(1) A – Corpus luteum, secretes large amounts of progesterone for the maintenance of endometrium.
(2) B – Ovum, released from the ovary after the completion of second meiotic division.
(3) C – Corpus luteum, secretes estrogen to maintain endometrium.
(4) D – Graafian follicle, containing primary oocyte in meiosis-I.

184. Which of the following statements is **incorrect** w.r.t. amniocentesis?
(1) It involves collection of small sample of amniotic fluid surrounding the foetus
(2) It is usually done in order to diagnose chromosomal abnormalities and other genetic disorders
(3) It involves withdrawal of sample of some of the placental tissue
(4) It helps in analysis of foetal cells.

185. Fill in the blanks and choose the **correct** option
For normal fertility at least __I__% the total sperms in a single ejaculation must have normal shape and size and __II__% of them must show vigorous motility.
(1) I – 40; II – 60
(2) I – 50; II – 50
(3) I – 30; II – 70
(4) I – 60; II – 40

CC-020 CC-020 CC-020 CC-020 CC-020 CC-020 CC-020
Aakash +DBuss
@NEETKART

SECTION-B

186. Identify the correct site of synthesis and pathway of flow of milk in a functional mammary gland and choose the correct option.

(1) Mammary alveoli → mammary ampulla → mammary duct → lactiferous duct

(2) Mammary alveoli → mammary duct → mammary ampulla → lactiferous duct

(3) Mammary alveoli → mammary duct → mammary lobe → mammary ampulla → lactiferous duct

(4) Mammary lobe → mammary duct → mammary ampulla → mammary alveoli → lactiferous duct

187. Which among the following structures is unpaired?

(1) Seminal vesicle

(2) Bulbourethral gland

(3) Mammary gland

(4) Prostate gland

188. The anterior portion of sperm head is covered by a cap-like structure called

(1) Nucleus

(2) Acrosome

(3) Middle piece

(4) Centriole

189. Chorionic villi and uterine tissue become interdigitated with each other and jointly form

(1) Embryo

(2) Umbilical cord

(3) Placenta

(4) Yolk sac

190. The embryo with 8 to 16 blastomeres stage is termed as morula which first develops into

(1) Zygote

(2) Placenta

(3) Trophoblast

(4) Blastocyst

191. **Assertion (A)** [illegible] is completed [illegible]
Reason (R) [illegible] ovum [illegible] division.
In the light of [illegible] correct option.

(1) Both [illegible] and [illegible] explanation [illegible]

(2) Both [illegible] the [illegible] explanation [illegible]

(3) Assertion is true [illegible] is false.

(4) Both Assertion and Reason are false statements.

192. Read the given statements.
Statement A: Leydig cells are located in interstitial spaces in testes.
Statement B: Sertoli cells of seminiferous tubules provide nourishment to the developing sperms.
Select the correct option.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

193. A lady is diagnosed to have both fallopian tubes completely blocked. Which of the following ART she cannot use to conceive a child?

(1) GIFT

(2) IUT

(3) IVF

(4) ICSI

194. Which of the following set of hormones is produced in females only during pregnancy?

(1) Estrogen, progesterone and prolactin

(2) hCG, hPL and relaxin

(3) hCG, hPL and cortisol

(4) hCG, hPL and progesterone

195. In vasectomised individuals

(1) Spermatogenesis is inhibited

(2) Testosterone synthesis is decreased

(3) Sexual desires are inhibited

(4) Sperms are prevented from reaching the male urethra

CC-020 CC-020 CC-020 CC-020 CC-020 CC-020
Aakash +Byju's
@NEETPDF

196. In the method of 'periodic abstinence' the couples avoid or abstain from coitus from day _____ of the menstrual cycle of 28 days when ovulation could be expected. Choose the option that fills the blank **correctly.**

(1) 5 – 6
(2) 1 - 10
(3) 10 – 17
(4) 22 - 28

197. Read the following statements.

A. Progestogen alone or in combination with estrogen can be used by females as injections or implants under the skin.

B. The mode of action of implants is similar to oral contraceptive pills but effective period is much shorter.

Choose the **correct** option.

(1) Both the statements are correct
(2) Only statement A is correct
(3) Only statement B is correct
(4) Both the statements are incorrect

198. Male germ cells undergo reductional division and eventually form spermatozoa. These germ cells line

(1) Rete testis
(2) Interstitial spaces
(3) Epididymis
(4) Seminiferous tubules

199. Lactational amenorrhea following parturition is an effective birth control method only upto a maximum period of

(1) 6 months
(2) 1 year
(3) 18 months
(4) 30 days

200. In males, __A__ stimulates the synthesis and secretion of hormones called __B__ from testes.

Choose the option that **correctly** fills the blank A and B.

(1) **A** – FSH, **B** – Estrogen
(2) **A** – LH, **B** – Androgens
(3) **A** – FSH, **B** – Androgens
(4) **A** – LH, **B** – Estrogen

@NEETADP
CC-020 CC-020 CC-020 CC-020
Aakash +BYJU'S
CC-020 CC-020 CC-020 CC-020